METRIC

MIL-DTL-87173B(USAF)
13 February 1998

SUPERSEDING
MIL-P-87173A(USAF)
23 June 1989

DETAIL SPECIFICATION

PROPELLANT, PRIMING FLUID, ALCM ENGINE, GRADE PF-1

Inactive for new design after 13 February 1998.

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of high-density hydrocarbon fuel. This specification was thoroughly reviewed as a part of acquisition reform. While most of the requirements were converted to performance terms, due to the military-unique nature of the product (see 6.1) and the need for compatibility with deployed systems, it was determined that not all requirements could be converted. The issuance of this specification as “detail” is not intended to constrain technology advances in future systems.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the *Department of Defense Index of Specifications and Standards (DoDISS)* and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASC/ENSI, Bldg 560, 2530 Loop Rd. West, Wright-Patterson AFB OH 45433-7101, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 9135

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:56Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-I-85470 Inhibitor, Icing, Fuel System, High Flash

STANDARDS

FEDERAL

FED-STD-791 Lubricants, Liquid Fuels, and Related Products; Methods of Testing

DEPARTMENT OF DEFENSE

MIL-STD-290 Packaging of Petroleum and Related Products

(Unless otherwise indicated, copies of the above specifications and standards are available from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the *DoDISS* cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the *DoDISS* are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS STANDARDS

ASTM D93	Test Methods for Flash Point by Pensky-Martens Closed Tester (DoD adopted)
ASTM D156	Test Methods for Saybolt Color of Petroleum Products (Saybolt Chromometer Method) (DoD adopted)
ASTM D240	Standard Test Method for Heat of Combustion of Liquid Hydrocarbon Fuels by Bomb Calorimeter (DoD adopted)
ASTM D381	Standard Test Method for Existent Gum in Fuels by Jet Evaporation (DoD adopted)
ASTM D445	Standard Test Method of Kinematic Viscosity of Transparent and Opaque Liquids (and the Calculation of Dynamic Viscosity) (DoD adopted)
ASTM D1298	Standard Test Method for Density, Relative Density (Specific Gravity), or API Gravity of Crude Petroleum and Liquid Petroleum Products by Hydrometer Method (DoD adopted)
ASTM D2276	Standard Test Methods for Particulate Contaminant in Aviation Turbine Fuels (DoD adopted)
ASTM D2382	Standard Test Method for Heat of Combustion of Hydrocarbon Fuels by Bomb Calorimeter (High-Precision Method) (DoD adopted)
ASTM D2386	Standard Test Method for Freezing Point of Aviation Fuels (DoD adopted)
ASTM D3241	Standard Test Method for Thermal Oxidation Stability of Aviation Turbine Fuels (JFTOT Procedures) (DoD adopted)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:56Z
Check the source to verify that this is the current version before use.

ASTM D3828	Standard Test Method for Flash Point by Setaflash Closed Tester (DoD adopted)
ASTM D4052	Standard Test Method for Density and Relative Density of Liquids by Digital Density Meter (DoD Adopted)
ASTM D4057	Standard Practice for Manual Sampling of Petroleum and Petroleum Products (DoD adopted)
ASTM D5006	Standard Test Method for Measurement of Fuel System Icing Inhibitors (Ether Type) in Aviation Fuels (DoD Adopted)
ASTM D5452	Standard Test Method for Particulate Contamination in Aviation Fuels by Laboratory Filtration (DoD Adopted)
ASTM D6045	Standard Test Method for Color of Petroleum Products by the Automatic Tristimulus Method
ASTM E29	Recommended Practice of Indicating Which Places of Figures are to be Considered Significant in Specified Limiting Values (DoD adopted)
ASTM E300	Sampling Industrial Chemicals, Practice for

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken PA 19428-2959; (610) 832-9500.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Except as otherwise specified herein, the priming fluid shall consist completely of a mixture of exo-tetrahydrodi(cyclopentadiene) and methylcyclohexane.

3.1.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2 Chemical and physical requirements. The chemical and physical requirements of the priming fluid shall conform to those listed in table I.

3.2.1 Additives. The additives listed shall be used in combination in amounts not to exceed those specified. Information concerning the type and amount of each additive used shall be made available when requested by Procuring Activity or User.

3.2.2 Antioxidants. The following active inhibitors shall be blended separately or in combination into the priming fluid in total concentration of 90 parts per million (minimum) to 110 parts per million (maximum), by weight, not including weight of solvent, in order to prevent the formation of polymeric oxidation products:

a. 2,6-di-tert-butyl-4-methylphenol

b. 6-tert-butyl-2,4-dimethylphenol

c. 2,6-di-tert-butylphenol

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:56Z
Check the source to verify that this is the current version before use.

3.2.3 Fuel system icing inhibitor. Fuel system icing inhibitor shall be used only at the request of the User. If requested by the User, the inhibitor shall conform to MIL-I-85470.

TABLE I. Chemical and physical requirements and test methods.

Property	Priming Fluid		ASTM Standards
	Minimum	Maximum	
Color, Saybolt	+25		D156 [4], D6045
Chemical composition, wt, percent			[1]
methylcyclohexane	10.0	15.0	
exo-tetrahydrodi(cyclopentadiene)	83.5	90.0	
other hydrocarbons		1.5	
Flash point, °C (°F)	16 (60)	27 (80)	D93 [4], D3828
Specific gravity, (60/60F)	0.900	0.930	D1298, D4052 [4]
Freezing point, °C (°F)		-54 (-65)	D2386
Viscosity, centistokes at -54 °C		40	D445
Net heat of combustion MJ/kg (Btu/lb)	42.1 (18,100)		D240, D2382
Thermal stability change in			D3241 [2]
pressure drop, mm Hg		10	
heater tube deposit visual rating		< Code 3	
Existent gum, mg/100 mL		5	D381
Particulate matter, mg/liter		1.0	D2276, D5452
Fuel system icing inhibitor, volume percent	0.10	0.15	D5006 [3]

[1] Test shall be performed using appendix A or appendix C.

[2] See 4.5.3 for ASTM D3241 test conditions and test limits.

[3] Use proper scale of refractometer.

[4] Referee Test Method

3.3 Performance. The priming fluid shall satisfy the performance requirements as specified in the tests of section 4, herein.

3.4 Workmanship. The priming fluid shall be visually free from undissolved water, sediment, or suspended matter and shall be clean and bright at the ambient temperature or at 21°C (70 °F), whichever is higher.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:56Z
Check the source to verify that this is the current version before use.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as quality conformance inspection (see 4.2).

4.2 Conformance inspection. Test for the acceptance of individual lots shall consist of tests for all requirements specified in section 3. Quality conformance inspection shall include the test requirement herein.

4.2.1 Inspection lot. For acceptance purposes, individual lots shall be examined as specified herein and subjected to tests for all requirements cited in section 3.

4.3 Inspection. Inspection shall be in accordance with FED-STD-791, Method 9601.

4.3.1 Inspection conditions. Requirements contained in table I are absolute as defined in ASTM E29, and shall not be subject to correction for test tolerances. If multiple determinations are made, results falling within any specified repeatability and reproducibility tolerances may be averaged. For rounding off of significant figures, ASTM E29, absolute method, shall apply to all tests required by this specification.

4.4 Sampling plans

4.4.1 Sampling. Each bulk or packaged lot (see 6.5) shall be sampled in accordance with ASTM E300. The samples shall be subjected to the examination specified in 4.5.1.1 and all tests specified under 4.5.2.

4.4.2 Sampling for inspection of filled containers. A random sample of filled containers shall be selected from each lot and shall be subjected to the examination of filled containers as specified in 4.5.1.3.

4.5 Methods of inspection

4.5.1 Examination of product

4.5.1.1 Visual inspection. Samples selected in accordance with 4.4.1 shall be visually examined for compliance with 3.4.

4.5.1.2 Examination of empty containers. Before filled, each empty unit container shall be visually inspected for cleanliness and suitability in accordance with ASTM D4057.

4.5.1.3 Examination of filled containers. Samples taken as specified in 4.4.2 shall be examined for conformance to MIL-STD-290 with regard to fill, closure, sealing, leakage, packaging, packing, and markings.

4.5.2 Chemical and physical tests. Tests to determine conformance to chemical and physical requirements shall be conducted in accordance with table I. The priming fluid shall pass all tests listed in table I. No additional testing shall be required. Requirements contained herein are not subject to corrections for test tolerances. If multiple determinations are made, results falling within any specified repeatability and reproducibility tolerances may be averaged. For rounding off of significant figures, ASTM E29 shall apply to all tests required by this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:56Z
Check the source to verify that this is the current version before use.

4.5.3 Thermal stability tests. The thermal stability test shall be conducted using ASTM D3241. The heater tube shall be rated visually (see appendix B).

4.5.3.1 Test conditions.

a. Heater tube temperature at maximum point: 260 °C (500 °F)

b. System pressure: 3.45 MPa (500 pounds per square inch gage)

c. Flow rate: 3.0 mL/minute

d. Test duration: 150 minutes.

4.5.3.2 Test results. The sample is acceptable if all the following criteria are met:

a. The maximum visual rating of the heater tube deposits is less than a code 3 (appendix B, B.3.6)

b. The visual rating of the heater tube shows neither peacock-type deposit (code P) nor abnormal-type deposits (Code A) (appendix B, B.3.6.3.1 and B.3.6.3.2)

c. The maximum differential pressure across the test filter does not exceed 10 millimeters of mercury.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The priming fluid covered by this specification is intended for use in the ALCM F107-WR-101 engine. This is a military-unique engine as is the priming fluid used to start it and has no commercial application. This priming fluid specification contains military-unique additives required for continual use of the priming fluid in current systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of *DoDISS* to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).

c. Quantity desired.

d. Type and size of the containers (see 5.1).

e. Applicable levels of packaging and packing (see 5.1).

f. Location, injection method, and type of additives, as required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:56Z
Check the source to verify that this is the current version before use.

6.3 Definitions

6.3.1 Bulk lot. A bulk lot consists of an indefinite quantity of a homogeneous mixture of material offered for acceptance in a single isolated container or manufactured in a single plant run through the same processing equipment, with no change in ingredient material.

6.3.2 Packaged lot. A packaged lot consists of an indefinite number of 208-liter (55-gallon) drums, or smaller unit packages of identical size and type, offered for acceptance and filled from the isolated tank containing a homogeneous mixture of material; or filled with a homogeneous mixture of material run through the same processing equipment, with no change in ingredient material.

6.3.3 Homogeneous product. A homogeneous product is defined as a product where samples taken at various levels of the batch tank are tested for the defining homogeneous characteristics and all values obtained meet the repeatability precision requirements for that test method.

6.4 Subject term (key word) listing.

Additive

Fuel

Hydrocarbon

Inhibitor

Thermal stability

Viscosity

6.5 Conversion of metric units. Units of measure have been converted to the International System of Units (SI) (Metric) in accordance with ASTM E380. If test results are obtained in units other than Metric or there is a requirement to report dual units, ASTM E380 or ASTM D1250, Volume XI/XII, should be used to convert the units.

6.6 Material Safety Data Sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force – 11

Preparing activity:
Air Force – 11

Review activity:
Air Force – 68

(Project 9135-F155)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:56Z
Check the source to verify that this is the current version before use.

APPENDIX A

METHOD OF TEST FOR ANALYSIS OF GRADE PF-1 PROPELLANT BY GAS CHROMATOGRAPHY

A.1 SCOPE

A.1.1 Scope. This method is intended for the analysis of PF-1 Propellant. The method provides for the quantitative determination of the two major constituents of PF-1 and for the characterization of each constituent.

A.1.2 Summary of methods. The sample is introduced into a gas chromatographic column and separated into its individual components. Conditions are selected to give complete resolution of the two major blending stocks. The detector signal is monitored continuously and the areas of individual peaks are used for computing the quantities of the components. Response factors are applied to give the fuel composition in true weight percent. The detailed composition of each major component is computed from the individual peak areas in the subgroup.

A.2 APPLICABLE DOCUMENTS. (The section is not applicable to this appendix.)

A.3 APPARATUS

A.3.1 Chromatograph. Any conventional gas chromatograph, having at least the following features, is acceptable.

A.3.1.1 Detector. Either a thermal conductivity (TCD) or flame ionization (FID) detector may be used. The detector shall be operated in such a manner that its linear dynamic range is not exceeded, and its linearity should be checked periodically. The detector shall be capable of continuous operation at 325°C or higher, and must be connected to the column in a manner that eliminates cold spots.

A.3.1.2 Temperature programmer. The instrument must be capable of reproducible temperature programming over the range of 50° to 300°C. The programming rate must be sufficiently reproducible that individual retention times do not vary by more than two percent. Any temperature program, including multilinear modes and intermediate periods of isothermal operation, is acceptable, provided that the column resolution, as defined in section A.5.2, is not compromised.

NOTE 1. If capillary-type columns are used for the analysis, it may not be necessary to program the oven above 200°C.

A.3.1.3 Sample inlet system. Either flash vaporization or on-column injection may be used. For flash vaporization, the sample inlet port must be maintained between 300° and 350°C, and well-conditioned septa must be used. If on-column injection is employed, provision must be made for programming the temperature of the full length of the column. A sample inlet splitter may be used in conjunction with capillary columns, if care is taken to ensure that a representative sample is delivered to the column.

NOTE 2. New septa should be allowed to condition overnight after being installed in order to minimize spurious peaks, and several blank runs should be made to purge the column of materials bleeding from the septum. Overnight conditioning can be eliminated by the expedient of storing a supply of septa in the column oven. However, at least one blank run should still be made following each septum change.

A.3.1.4 Recorder. Any potentiometric recorder compatible with the chromatograph may be used. The full-scale response time should be two seconds or less, and the gain and damping controls should be adjusted to provide optimum response and minimum noise.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:56Z
Check the source to verify that this is the current version before use.

A.3.1.5 Column. Either packed or open-tubular (capillary-type) columns may be used, provided the minimum resolution, specified in section A.5.2, is maintained. The column should be designed for maximum thermal stability and minimum bleed, since the temperature program is likely to run as high as 275 °C for packed column and 200 °C for capillary columns. Dual-column compensation is required if column bleed results in excessive baseline drift at high temperatures.

A.3.1.6 Integrator. An electric integrator is required for highest accuracy and precision. Data acquisition and reduction by an on-line computer, programmed for gas chromatography, is convenient and may be used if such capability is available.

A.3.1.7 Microsyringe. A microsyringe, capable of delivering sample volumes from 0.5 to 5.0 microliters, is required.

NOTE 3. The widely used 0 - 10 microliter syringe, Hamilton Model 701N or its equivalent, is recommended. These syringes are routinely available from most instrument manufacturers and chromatography supply houses.

A.4 REAGENTS AND MATERIALS

A.4.1 Stationary-phase. The stationary phase should be one of the familiar non-polar silicones. The following silicones have been used successfully:

Silicone SE-30

Silicone UCW-982

Silicone OV-101.

NOTE 4. UCW-982 is recommended because it exhibits less column bleed at high temperatures than the other silicones. However, any of the listed phases are satisfactory, provided baseline drift at high temperature is minimized by conditioning or by dual column compensation.

A.4.2 Carrier gas. Either helium or nitrogen is suitable, although nitrogen is known to give somewhat higher column efficiencies at lower flow rates, and is more economical. Helium must be used with thermal conductivity detectors. The flow must be maintained constant to within one percent throughout the entire range of operating temperatures by means of constant flow controllers.

A.4.3 Hydrogen. High purity, for flame ionization detectors

A.4.4 Air. Dry, filtered, and oil-free, for flame ionization detectors

A.4.5 Exo-tetrahydrodi(cyclopentadiene). (99 percent purity), hereafter referred to as C-10

NOTE 5. High purity samples of C-10 may be obtained by submitting a written request to AFRL/PRSF, 1790 Loop Rd. N., Wright-Patterson AFB OH 45433-7103.

A.4.6 Methylcyclohexane. (99 percent purity) hereinafter referred to as MCH

A.4.7 n-Pentane, n-hexane, or carbon disulfide. Chromatographic purity

A.4.8 Qualitative calibration mixture. A series of known hydrocarbons is used to calculate column resolution. Prepare a mixture containing about one percent (by volume) each, in pentane, hexane, or carbon disulfide of at least eight normal alkanes and six cycloalkane derivatives selected from highly pure (minimum 99 percent) samples which have boiling points in the 70°C-250°C range. The mixture must contain MCH, n-dodecane, and n-tetradecane.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:56Z
Check the source to verify that this is the current version before use.

A.4.9 Quantitative calibration mixture. Prepare several calibration mixtures by carefully weighing quantities of MCH and C-10 into sample vials or other suitable closed containers. Handling and weighing should be performed carefully so the weight of each component is accurately known to within one percent (relative).

NOTE 6. The calibration mixtures shall be of varying compositions in order to confirm the linearity of the detector and the validity of the calibration; response factors should be constant and independent of the actual compositions. The compositions should approximate the range defined in the PF-1 specification but it is helpful to prepare the samples so these limits are occasionally exceeded.

NOTE 7. Weighing may be done by difference into any convenient containers, provided care is taken to introduce the materials quickly and thereby avoid loss by evaporation. Since MCH has appreciable vapor pressure under ambient conditions, it should be weighed last. The containers should be leak-tight, and stored in a cold box (preferably a freezer compartment). Properly prepared and stored samples are stable for several months.

A.5 PREPARATION OF APPARATUS

A.5.1 Column. Any conventional method of conditioning and installing the column is satisfactory, provided the system is free of gas leaks and the column resolution meets the minimum requirement specified in A.5.2.

A.5.2 Column resolution. Run a sample of the qualitative calibration mixture and measure the retention times and widths of the n-dodecane and n-tetradecane peaks as shown on figure A1. Resolution is defined as the difference in retention times for n-C_{12} and n-C_{14} divided by their average width:

$$R = \frac{2(RT_{14} - RT_{12})}{W_{14} + W_{12}}$$

Resolution determined in this way must be at least 5.0 and can be optimized by varying the temperature program and the carrier flow rate. If resolution cannot be increased to at least 5.0 by optimizing these parameters, it is necessary to use a different column.

A.5.3 Chromatograph. Place in service according to the manufacturer's specifications; give particular attention to flow rates through the detector.

A.6 PROCEDURE

A.6.1 Establish a suitable sample volume by making several trial runs with the qualitative calibration mixture. For FID instruments, the volume may be between 0.5 and 1.0 microliter; for TCD instruments, volumes from 1.0 to 5.0 microliters have been found satisfactory.

NOTE 8. To establish the optimum temperature program, it is necessary to begin the run at a low enough temperature to allow complete elution of the solvent before the appearance of the MCH peak. If the area is being monitored by an electronic integrator or computer, a time delay should be entered to inhibit integration of the solvent peak(s).

A.6.2 Dilute an aliquot of each quantitative calibration mixture at least 1:10 (preferably 1:20) with either pentane, hexane, or carbon disulfide. Mix thoroughly and inject a measured volume into the chromatograph, simultaneously activating the temperature program, the integrator, and the recorder.

A.6.3 Integrate the area under the chromatogram continuously until the final peak has been eluted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:56Z
Check the source to verify that this is the current version before use.

A.6.4 Compute a response factor for MCH as described in A.7. Repeat the procedure at least three times for each calibration sample and find the mean value for each response factor. Then dilute the PF-1 acceptance sample and run in the same way as the calibration standards, and determine its composition as described in A.7.

NOTE 9. For MCH and C-10, extra peaks which are within plus or minus ten percent of the retention time for MCH or C-10, respectively, shall be considered part of the MCH or C-10 peak areas. If commercial grade MCH was used to blend the PF-1 acceptance sample, appreciable amounts of heptane and toluene may appear as resolved peaks near the MCH peak. These peak areas should be added, uncorrected, to the area of the MCH peak. Assuming unit relative response factors for these materials does not seriously affect the accuracy of the method.

A.7 CALCULATIONS

A.7.1 C-10 is used in each case as an internal standard, and is considered to have a response factor of 1.000. After analysis with each calibration standard, group the peak areas which correspond to the MCH and calculate the relative response for MCH by 70.2.

$$K_{MCH} = \frac{wt(MCH)}{wt(C\text{-}10)} \quad X \quad \frac{area\ (C\text{-}10)}{area\ (MCH)}$$

For each PF-1 acceptance sample, apply the calculated MCH response factor to each MCH peak area, and sum the products to obtain a corrected MCH area. In the following equation, all hydrocarbons other than those contained in the MCH and C-10 subgroups are considered impurities. A response factor of 1.000 is used for the impurities. Calculate the weight percent of each PF-1 component as follows:

$$\%MCH = \frac{K_{MCH}\ A_{MCH}}{K_{MCH}\ A_{MCH} + A_{C\text{-}10} + A_{Impurities}} \quad X\ 100$$

$$\%C\text{-}10 = \frac{A_{C\text{-}10}}{K_{MCH}\ A_{MCH} + A_{C\text{-}10} + A_{Impurities}} \quad X\ 100$$

$$\%\ Other\ hydrocarbons = 100 - (\%MCH + \%\ C\text{-}10)$$

.

A.8 PRECISION

A.8.1 Precision. Detailed precision studies have not been performed for this method. However, limited results indicate the method is capable of the following precision.

A.8.2 Repeatability. Replicate analyses, performed under identical conditions in the same laboratory, should yield results which do not vary more than one percent (absolute) from the mean value for each component.

A.8.3 Reproducibility. When the same sample is analyzed in different laboratories according to this method, the results for any component should not differ by more than 1.5 percent (absolute).

A.8.4 Accuracy. Back-analysis of a calibration standard should give results which agree with the known values to within 0.5 percent (absolute).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:56Z
Check the source to verify that this is the current version before use.

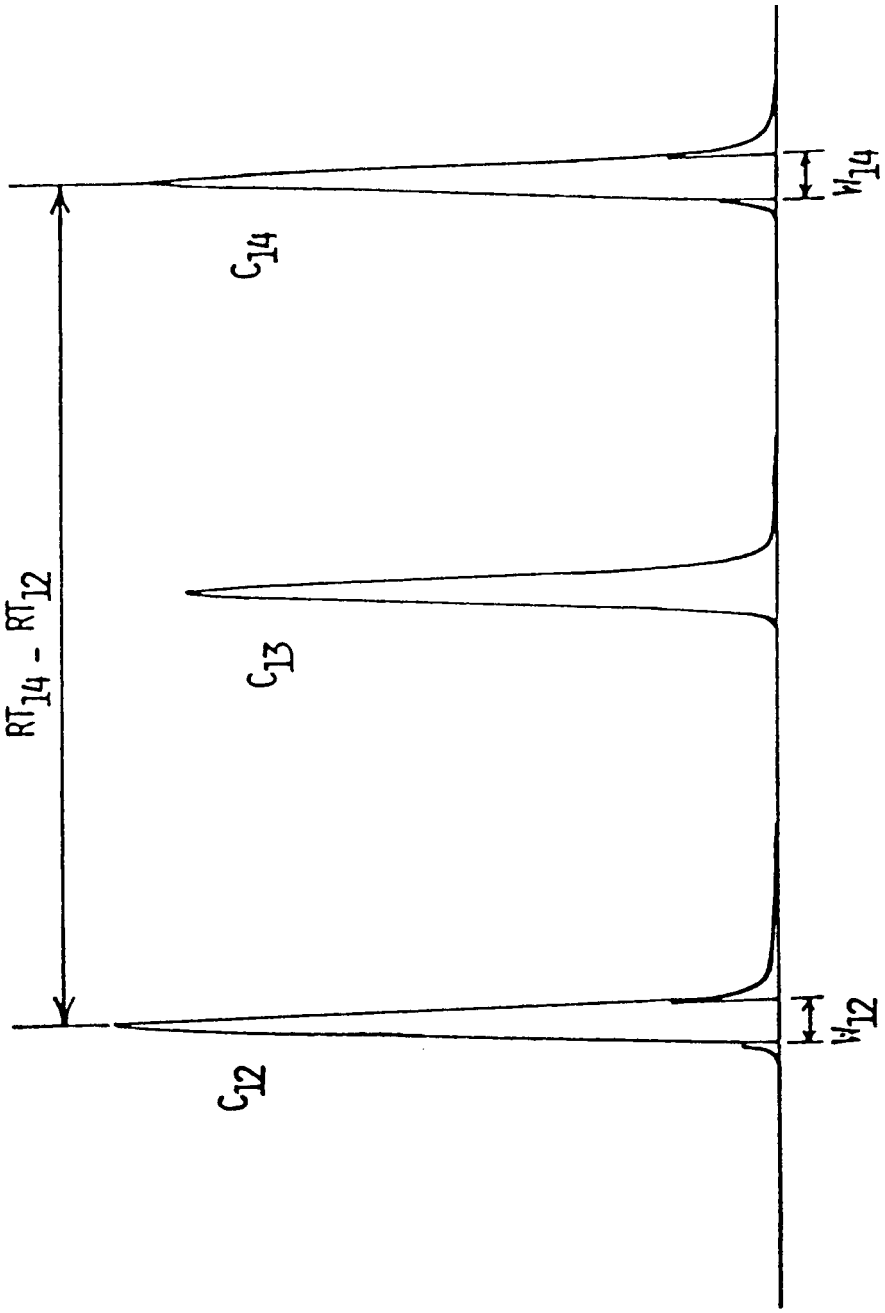


FIGURE A1. Calculation of column resolution.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:56Z
Check the source to verify that this is the current version before use.

APPENDIX B

HEATER TUBE DEPOSIT RATING

B.1 SCOPE

B.1.1 Scope. This appendix provides instructions to conduct and read the heater tube deposit. This appendix is a mandatory part of the specification.

B.2 APPLICABLE DOCUMENTS. (This section is not applicable to this appendix.)

B.3 VISUAL TEST METHOD

B.3.1 Snap the upper end of the heater tube into the clamp of the adapter for the heater tube.

B.3.2 Push the heater tube against the stop of the adapter tube.

B.3.3 Slide the adapter with the heater tube over the guide rod into the tuberator equipped with a magnifying glass assembly.

B.3.4 Insert the ASTM color standard into the tuberator.

B.3.5 Rotate the adapter and position the heater tube so the side with the maximum deposit is visible.

B.3.6 Within 30 minutes after completion of the test, visually examine the heater tube in a tuberator. The entire portion of the test section between the bottom shoulder and the top shoulder of the heater tube test section shall be carefully examined using a magnifying glass in conjunction with the tuberator for any signs of discoloration, scratches, or other visually identified defects. When an area of the tube corresponds visually to an ASTM color standard, that color standard code number shall be recorded. If the area being rated has a color between two adjacent color standards, it shall be rated as the lighter (lower number) color standard. (NOTE: It is important that all light bulbs in the tuberator are functioning, as a change in light intensity can shift the rating significantly.)

(NOTE: The person rating the tube should have normal ability to distinguish between colors; i.e., he should not be color blind.)

B.3.6.1 In rating the heater tube, the darkest deposits govern and the code number representative of the darkest section, rather than the average deposit, shall be reported.

B.3.6.2 If a spot or streak is found on the heater tube, it shall be carefully examined under various lighting conditions using a magnifying glass to determine if it is a deposit, a scratch, or tube defect; (note that the tube defects should have been found during the pretest inspection of the tube). If the spot or streak is determined to be a scratch or tube defect, it shall be disregarded. If the spot or streak is a deposit, it shall be rated against the ASTM color standards, if larger in area than about 0.025 sq cm (0.004 square inch); i.e., approximately 1.5 mm x 1.5 mm ($^1/_{16}$ inch x $^1/_{16}$ inch) square or an equivalent area. However, a streak deposit shall be ignored if less than 0.8 mm ($^1/_{32}$ inch) wide, regardless of length. Note that the tube section is about 3 mm ($^1/_8$ inch) in diameter; thus, a 1.5 mm- ($^1/_{16}$ inch-) wide spot is half the diameter of the tube test section and a 0.8 mm- ($^1/_{32}$ inch-) wide streak is one fourth the diameter of the tube test section.

B.3.6.3 If the heater tube has deposits which do not match the Color Standards, the following criteria shall be used.

B.3.6.3.1 If the deposit has peacock (rainbow) colors, rate this as Code P (P for peacock). If some portion of the deposit does match the Color Standard, it shall be rated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:56Z
Check the source to verify that this is the current version before use.

B.3.6.3.2 Abnormally-colored deposits (for example, blue or gray) shall be rated code A (A for abnormal color).

B.3.6.3.3 When the overall tube rating is reported, record the rating of the maximum deposit that matches the Color Standards plus P or A if the tube contains deposits which do not match the Color Standards. If the tube contains only P or A deposits, report only the appropriate letter(s); do not try to assign a numerical rating to a P or A deposit. Examples of how the rating procedure is to be used are given below:

EXAMPLE 1: The darkest deposits on the heater tube match color standard 3. Also present are peacock colors. Thus, the overall tube rating to be reported is 3P.

EXAMPLE 2: The heater tube has maximum deposits which fall between Color Standards 2 and 3 and has no peacock or abnormal colors. The total tube rating is 2.

EXAMPLE 3: The heater tube matches Color Standard 1 except for an abnormal deposit that does not match the ASTM Color Standards. The overall tube rating to be reported is 1A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:56Z
Check the source to verify that this is the current version before use.

APPENDIX C

METHOD OF TEST FOR ANALYSIS OF GRADE PF-1 FUEL BY GAS CHROMATOGRAPHY

C.1 SCOPE

C.1.1 Scope. This method is intended for the analysis of PF-1 Propellant. The method provides for the quantitative determination of the two major constituents of PF-1 and for the characterization of each constituent.

C.1.2 Summary of methods. The sample is introduced into a gas chromatographic column. The detector signal is monitored continuously and the areas of individual peaks are used for computing the quantities of the methylcyclohexane, exo-tetrahydrodi(cyclopentadiene), and other hydrocarbon impurities. The detailed composition of the fuel components are computed from the individual peak areas in the subgroup.

C.2 APPLICABLE DOCUMENTS. (The section is not applicable to this appendix.)

C.3 APPARATUS

C.3.1 Chromatograph. Any conventional gas chromatograph, having at least the following features, is acceptable.

C.3.1.1 Detector. A flame ionization detector (FID) must be used. The detector shall be operated in such a manner that its linear dynamic range is not exceeded, and its linearity should be checked periodically. The detector shall be capable of continuous operation at 300°C or higher, and must be connected to the column in a manner that eliminates cold spots.

C.3.1.2 Temperature programmer. The instrument must be capable of reproducible temperature programming over the range of 115° to 250°C. The programming rate must be sufficiently reproducible that individual retention times do not vary by more than two percent. The temperature program consists of an initial 12 minutes at 115°C after injection, followed by a programmed temperature rise of 10°C per minute up to 250°C, and culminating with 10 minutes at 250°C (total run time = 35.5 minutes).

C.3.1.3 Sample inlet system. Flash vaporization shall be used with the sample inlet port maintained at 250°C, and fitted with a well-conditioned septa. A sample inlet splitter set at ~100:1 shall be used in conjunction with the capillary column described in C.3.1.5.

NOTE 1. New septa should be allowed to condition overnight after being installed in order to minimize spurious peaks, and several blank runs should be made to purge the column of materials bleeding from the septum. Overnight conditioning can be eliminated by the expedient of storing a supply of septa in the column oven. However, at least one blank run should still be made following each septum change.

C.3.1.4 Recorder. Any potentiometric recorder compatible with the chromatograph may be used. The full-scale response time should be two seconds or less, and the gain and damping controls should be adjusted to provide optimum response and minimum noise.

C.3.1.5 Column. A 60 meter J&W DB-1 capillary column of 0.25 mm ID and 0.25μ coating, or equivalent, shall be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:56Z
Check the source to verify that this is the current version before use.

C.3.1.6 Integrator. An electric integrator is required for highest accuracy and precision. Data acquisition and reduction by an on-line computer, programmed for gas chromatography, is convenient and may be used if such capability is available.

C.3.1.7 Microsyringe. A microsyringe or auto sampler, capable of delivering sample volumes from 0.5 to 5.0 microliters, is required.

C.4 REAGENTS AND MATERIALS

C.4.1 Stationary-phase. The capillary column is described in C.3.1.5.

C.4.2 Carrier gas. Nitrogen gas with 1.406 Kg/cm^2 (20 psi) head pressure shall be used.

C.4.3 Hydrogen. High purity, for flame ionization detectors

C.4.4 Air. Dry, filtered, and oil-free, for flame ionization detectors

C.4.5 Exo-tetrahydrodi(cyclopentadiene). (99 percent purity), hereafter referred to as JP-10

NOTE 2. High-purity samples of JP-10 may be obtained by submitting a written request to AFRL/PRSF, 1790 Loop Rd. N., Wright-Patterson AFB OH 45433-7103.

C.4.6 Methylcyclohexane. (99 percent purity) hereinafter referred to as MCH

C.4.7 n-Pentane, n-hexane, or carbon disulfide. Chromatographic purity.

C.4.8 Qualitative calibration mixture. A series of known hydrocarbons is used to calculate column resolution. Prepare a mixture that contains about one percent (by volume) each, in pentane, hexane, or carbon disulfide of at least two normal alkanes and two cycloalkane derivatives selected from highly-pure (minimum 99 percent) samples which have boiling points in the 70°C-250°C range. The mixture must contain MCH, n-dodecane, and n-tetradecane.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:56Z
Check the source to verify that this is the current version before use.

C.5 PREPARATION OF APPARATUS

C.5.1 Column. Any conventional method of conditioning and installing the column is satisfactory, provided the system is free of gas leaks and the column resolution meets the minimum requirement specified in C.5.2.

C.5.2 Column resolution. As a final check on chromatographic equipment and settings, run a sample of the qualitative calibration mixture and measure the retention times and widths of the n-dodecane and n-tetradecane peaks as shown on Figure A1. Resolution is defined as the difference in retention times for n-C12 and n-C14 divided by their average width:

$$R = \frac{2(RT_{14} - RT_{12})}{W_{14} + W_{12}}$$

Resolution determined in this way must be at least 5.0 and can be optimized by varying the temperature program and the carrier flow rate. If resolution cannot be increased to at least 5.0 by optimizing these parameters, it is necessary to use a different column and follow the general procedure of appendix A.

C.5.3 Chromatograph. Place in service according to the manufacturer's specifications; give particular attention to flow rates through the detector.

C.6 PROCEDURE

C.6.1 Sample volume shall be approximately 0.5 microliter. This sample is injected neat, which means no dilution in a solvent.

C.6.2 Inject a measured volume of sample into the chromatograph while simultaneously activating the temperature program, the integrator, and the recorder.

C.6.3 Integrate the area under the chromatogram continuously until the final peak has been eluted. (Area under JP-10 peak = $A_{JP\text{-}10}$; Area under MCH peak = A_{MCH}; Area under other peaks = $A_{Impurities}$ in C.7.1.)

NOTE 3. When calculating area under JP-10, include area under peaks with $\pm$10% retention time of JP-10 if they do not contribute more than 20 percent to the total C-10 peak areas.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:56Z
Check the source to verify that this is the current version before use.

C.7 CALCULATIONS

C.7.1 For each PF-1 acceptance sample, calculate the weight percent of JP-10 and MCH as follows:

$$\%\text{JP-10} = A_{\text{JP-10}} \frac{A_{\text{JP-10}}}{A_{\text{JP-10}} + A_{\text{MCH}} + A_{\text{impurities}}} \text{ X } 100$$

$$\%\text{MCH} = A_{\text{JP-10}} \frac{A_{\text{MCH}}}{A_{\text{JP-10}} + A_{\text{MCH}} + A_{\text{impurities}}} \text{ X } 100 \quad .$$

In the above equation, all hydrocarbons other than those contained in the JP-10 and MCH subgroup are considered impurities. A response factor of 1.000 is used for the impurities.

C.8 PRECISION

C.8.1 Precision. Detailed precision studies have not been performed for this method. However, limited results indicate the method is capable of the following precision.

C.8.2 Repeatability. Replicate analyses, performed under identical conditions in the same laboratory, should yield results that do not vary more than one percent (absolute) from the mean value for each component.

C.8.3 Reproducibility. When the same sample is analyzed in different laboratories according to this method, the results for any component should not differ by more than 1.5 percent (absolute).

C.8.4 Accuracy. Back-analysis of a calibration standard should give results that agree with the known values to within 0.5 percent (absolute).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:56Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 7. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, and 6.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waiver any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER **MIL-DTL-87173B(USAF)**	2. DOCUMENT DATE (YYMMDD) **980213**

3. DOCUMENT TITLE
PROPELLANT, PRIMING FLUID, ALCM ENGINE, GRADE PF-1

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	e. DATE SUBMITTED *(YYMMDD)*

7. PREPARING ACTIVITY

a. NAME ASC/ENSI Air Force Code 11	b. TELEPHONE (Include Area Code) (1) Commercial (937) 255-6295	(2) AUTOVON 785-6295
c. ADDRESS *(Include Zip Code)* 2530 Loop Road West Wright-Patterson AFB OH 45433-7101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD FORM 1426, OCT 89 (EF-V1) PREVIOUS EDITIONS ARE OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:56Z
Check the source to verify that this is the current version before use.